

Assignment P5

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1 QUESTION 1 - COMPUTER SCIENCE PROMPT

1.1 Positive Effect of OMSCS

One positive effect of OMSCS is that it gives opportunities to people to pursue a masters degree who otherwise might not have, due to the cost and asynchronous structure. For example, researchers found that the average applicant to the in-person program was a 24-year-old international student, while the average applicant to OMSCS was a 34-year-old American (Goodman, Melkers, and Pallais, 2019). These are very different demographics; the in-person program is for people who can study full-time, while the OMSCS program is for people who typically work full time and complete coursework after work or on weekends.

This is a positive effect because it contributes to diversity of thought in the computer science community. It takes a certain level of privilege to have the resources and support to move to a new city and forego an income for two years while they study full time. Older students may already have established roots in their current city or town, may not want to leave their jobs, or may not be able to afford to leave their jobs (e.g. if they are supporting a family). They have different life experiences and have different perspectives that they can bring to the difficult problems in computer science. There are many instances of biases affecting important problems in computer science, such as Amazon's recruiting algorithm that was biased against women (Kodiyan, 2019). Including students with diverse backgrounds and life experience can help prevent biases like these.

1.2 Negative Effect of OMSCS

One negative effect of OMSCS is that because it is not based on a physical campus, it does not afford international students the same opportunities to come to the U.S. that in-person program does. Students in the in-person program can be sponsored for a visa to come to the U.S. by Georgia Tech, but OMSCS students cannot (OMSCS, 2021). This is a problem because, similar to the section above, diversity of thought is very important for the computer science community so that major problems are solved with many perspectives in mind. The U.S. is a major technology hub, so international students may want to study in

the U.S. so that they can gain experience doing an internship or co-op at a local tech company while they study. Meanwhile, U.S. companies would gain new perspectives from their international hires on how to solve interesting problems and prevent U.S.-centered biases. These students' options are therefore to either pay more money for the in-person program or stay in their home country and not have their perspectives included in development of these new technologies.

1.3 Potential Redesign

One way OMSCS could be restructured to keep the positive effect, but limit the negative effect of the lack of international student visas, is by creating relationships or even formal programs with U.S.-based tech companies. For example, a company like Microsoft could partner with OMSCS to offer specific internship roles to international OMSCS students. This would mean that international students would still benefit from the online price tag of OMSCS, and they could complete their internship in any U.S. city because the program is asynchronous, plus they would get sponsored for a visa by the tech company instead of by Georgia Tech. This would not affect any federal regulations that Georgia Tech may be under for the visa situation, and international students would still benefit from receiving a visa and working the U.S., plus the tech company would benefit from the different perspectives of the international students.

2 QUESTION 2 - COMPUTER SCIENCE PROMPT

2.1 Chosen Area

The area that I have chosen to discuss is changing your own name on a plane ticket, for example if your legal name changed since the time you booked the flight (in the case of marriage, divorce, etc.). I will use Southwest Airlines as the example, though I am sure the process is similar for other airlines as well.

2.2 Stakeholders

2.2.1 Users/Passengers on Southwest Airlines

The passengers on Southwest Airlines are the first stakeholders. Their motivation is to update their legally changed name on their ticket as easily as possible (e.g. on the Southwest website) so that they can board the plane without issue.

2.2.2 Southwest Airlines

Southwest Airlines as a whole has conflicting motivations. Their first motivation is to make their passengers happy to get good reviews, so they want the process of updating the passengers' names to be as easy as possible. They also want to save time and money by not tying up their customer service agents in a lot of inquiries around this topic. However, their biggest motivation is to not get sued, fined, or in some other way punished by the U.S. federal government for not following regulations on proper documentation of passengers' information.

2.2.3 The U.S. Federal Government, in particular TSA (Transportation Security Administration)

The U.S. Federal Government, and in particular TSA, want to have well-documented flight passenger information documented at all times. This is for safety reasons; for example, if a plane goes down they need to know who all was on board. This is also for national security reasons; if there is a terrorist attack, for example, they need to know the identify of the terrorists and victims. Therefore, passengers cannot be allowed to change their names constantly, and they need to have proper documentation of any name changes.

3 MOTIVATIONS AND THEIR AFFECT ON DESIGNS

If political motivations were not a factor, then the easiest way for a passenger to update their name would be to log in to their account either on the mobile app or the website, find their flight information, click "edit", and edit their name right there in the portal. Unfortunately, that option does not exist, which leads passengers to the Southwest website's customer service document on how they are supposed to edit their name. This website details that passengers must contact Southwest through customer service channels or social media, then they must send a picture of their updated government ID or documents (such as a marriage license) (Southwest, 2020).

If Southwest had less stringent regulations imposed on them by the federal government, they probably could just have users create a customer service request and upload the image of the documentation in the portal. Though it is not explained anywhere why this is not possible, my assumption is that they need users to communicate with a customer service agent to verify the passenger's identity, which is probably a stipulation in the federal regulation. The fact that

Southwest does provide a variety of ways customers can contact customer service shows that they are trying to abide by federal regulations while being as flexible as possible for their customers and make them happy.

Finally, the federal government's motivation to have strict documentation on all passengers on all flights is the driving factor behind why passengers cannot simply edit their information in their portal. This is also why passengers must show proof of a government-issued legal document that displays their name change, and not just take their word for it or show something less legally binding like a utility bill. When the passenger arrives at the airport, they will be required to show their government-issued ID to TSA as a further check that the passenger did not falsify this name change on their account.

4 QUESTION 3

4.1 "Flexible and Mindful Self-Tracking: Design Implications from Paper Bullet Journals" by Ayobi, Amid and Sonne, Tobias and Marshall, Paul and Cox, Anna L.

4.1.1 *Summary*

The authors of this study were interested in the rise of paper bullet journals, despite the fact that automated digital journals are also available. The authors studied many posts about bullet journals on Instagram and found that paper journaling is about more than automation; it also has a visual and social component (Ayobi et al., 2018).

The authors first went on Instagram to compile an assortment of 386 photos by searching for tags such as bulletjournal and habittracker. They found that users combined various aspects of different journals to meet their needs, customized the journals with creative aspects (colors, stickers, etc.), reflected on their journals regularly, and shared their journals on Instagram as part of a social component (Ayobi et al., 2018).

They argue that technology can help journals if it augments (instead of replaces) paper journals. For example, technology could be used to create digital templates that can be printed out for paper journaling. Also, social media such as Instagram is already a digital extension because it adds a social component to the journals. They also found that manually recording in the journals helps users

be creative, reflect, and meditate, so any added technology needs to focus on those aspects of journaling instead of just the need to record certain information (Ayobi et al., 2018).

4.1.2 Commentary

I wanted to read this paper because I also read about paper and digital journals at the beginning of the year in preparation for a new semester of school. I found that I was having trouble balancing work and school, and wondered if journaling might help keep me on track. I read a lot about bullet journaling and saw plenty of examples of how people use it in their daily lives. However, unlike the users in this study, I did not enjoy the bullet journal method. While it is visually pleasing and did encourage me to keep up with certain habits, I found that it was incredibly time consuming and ended up becoming a chore instead of a helpful support system. It also added to my guilt if I didn't spend time on it for a few days (why did I spend so much time on the journal if I wasn't going to continue with it?). I was interested to see if the authors of this paper addressed people like me who did not benefit from paper journals, but they did not. Overall though, I am glad that I read the paper and I learned more about how paper journals could be augmented by technology.

4.2 "A Stalker's Paradise': How Intimate Partner Abusers Exploit Technology" by Freed, Diana and Palmer, Jackeline and Minchala, Diana and Levy, Karen and Ristenpart, Thomas and Dell, Nicola

4.2.1 Summary

The authors of this study wanted to understand how technology is being used for IPV (Intimate Partner Violence). They interviewed 89 people, 39 of whom were victims of IPV and 50 of whom worked in roles assisting victims of IPV, such as social workers. Protecting victims in this case is difficult because abusers have access to so much information about their victims and may even have full access to victims' accounts (Freed et al., 2018).

One threat model that the authors describe is a "UI-bound adversary", or someone who is an authenticated user on the victim's account but performs actions that harm the victim, such as posting embarrassing information. They note that abusers can also use legitimate applications such as "Find My Friends" to stalk their victims. The four main categories of abuse that the authors describe are:

using their ownership of devices or accounts, compromising accounts, making harmful posts, or exposing information about victims (Freed et al., 2018).

The authors discuss how this can be an important sector of HCI: how to limit usability for abusers while still maintaining usability for legitimate users or victims. They make a few suggestions, like requiring two different passwords, one for unimportant information and one for potentially dangerous information (like locations). They also suggest adding IPV safety reviews to the interface design cycle (Freed et al., 2018).

4.2.2 Commentary

I selected this paper because I have recently been reading about and watching documentaries about domestic violence, and I was interested to see how HCI applies to domestic violence. There was a lot of really interesting information in this article, and definitely some considerations that I had not thought of, like how can interface designers protect authorized users from other authorized users on the same account. That sounds like an incredibly difficult task and not something that can easily be solved. However, learning from victims, incorporating safety reviews, and considering the dangerous applications of the various features of the application all sound like a good place to start.

5 QUESTION 4

5.1 "A Robot Barista Comments on its Clients: Social Attitudes Toward Robot Data Use" by Hedao, Samarendra and Williams, Akim and Wadgaonkar, Chinmay and Knight, Heather

This paper was from the International Conference on Human-Robot Interaction.

5.1.1 Summary

The authors of this study wanted to understand how robots' comments were perceived by humans in "gray area" settings like coffee shops (public areas where people might still have private conversations, like job interviews). They primarily looked at three variables: the sentiment of what the robot said (positive, neutral, or negative), the data source (database scan, conversation interpretation, body language analysis, or general coffee shop banter), and to whom the robot addressed (both people, one person, or to one person but about the other person) (Hedao et al., 2019).

The authors then created 288 scripts based on all of these possibilities (e.g. positive sentiment, database scan, to both people). There were also different situations, like job interviews or two people meeting to see if they wanted to be roommates. They hired two actors to act out the different scripts so that they wouldn't have to worry about real people's information being put at risk. They also videotaped these sessions and then asked over 4,000 online participants survey questions about what they thought (Hedao et al., 2019).

The survey results found that participants' impression of the interaction was largely based on the sentiment of what the robot said and the data source. Participants did not mind body language analysis but did mind conversation analysis and database scans. Finally, they thought it was impolite if the robot addressed one person but not the other. Overall, this study gives insight in how robots would be perceived by customers if they were used in more mainstream areas like coffee shops (Hedao et al., 2019).

5.1.2 Commentary

I found this paper very interesting because I had never considered how the robot would interact with customers if they were used in places like coffee shops. I assumed the robot would be extremely neutral, like only ask what the shopper wanted to order, or answer questions about the menu. The idea that the robot would try to communicate with people like a human employee would seem honestly a little odd to me. It is also interesting how the participants felt about body language analysis and the politeness of only one person being addressed versus both people. Personally, I would feel uncomfortable if a robot was analyzing my body language and I wouldn't want the robot to address either person about personal matters, so it is not impolite to me if a robot only spoke to one person. Human-robot interactions are still very novel (to me at least!) so I enjoyed learning more about how others perceive the relationship and how it differed from my own opinions.

5.2 "Exploring the Future Experience of Automated "Valet Parking" - a User Enactment" by Neuhaus, Robin and Lenz, Eva and Borojeni, Shadan Sadeghian and Hassenzahl, Marc

This paper was from the International Conference on Automotive User Interfaces and Interactive Vehicular Applications.

5.2.1 *Summary*

In this study, the authors wanted to understand how users would perceive the situation of a self-driving car parking itself in a parking garage. They were interested in how users experienced this situation and if different levels of feedback from the car affected the experience. They tried out three different levels of feedback: a bare minimum option, a middle-of-the-road option, and an option with as much feedback as possible (including a video of the car's actions). The middle and high feedback options also give users more options, such as choices for which garage to park in. This gives the user more control over the situation (Neuhaus et al., 2019).

For the study itself, these features did not exist yet, so they simulated it with a real person driving the car while the participants still acted like the car was driving itself. Users reported that parking in a parking garage already gives them stress and anxiety (small spaces, many people trying to get in and out at the same time, etc.), so this relieved some of that anxiety but added new fears. Some users were afraid the car would be stolen, or were worried about where it parked. Users reported enjoying the "middle of the road" option more than the bare minimum option because it gave them more peace of mind. However, some enjoyed the high feedback option because it gave them even more security, while others thought it was too much information and added back some stress of having to worry about the car (Neuhaus et al., 2019).

5.2.2 *Commentary*

I read an article a few years ago about how self-driving cars would change how parking garages are treated in the future; self-driving cars could drive further to park at cheaper garages, meaning that it could bring the cost down for garages overall. Therefore, I was interested to read this article and see if it investigated a similar topic. This article investigated a different aspect of self-driving cars parking in garages, but it was similarly very interesting. I agree that parking in parking garages is stressful, and allowing the car to park itself alleviates some of that stress. I think it is interesting that the high-feedback option was added back some of the stress for some users, because now they felt responsible for the car even though they were not in control of the car. Overall, I enjoyed reading about this experiment and seeing how this exciting technology interacts with common human worries and stressors.

6 REFERENCES

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